

Laeq Aslam

Energy & Urban Climate | Physics-Informed Time-Series Forecasting | Edge Deployment

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Professional Summary

PhD candidate in Control Science, developing **physics-informed AI** to advance the UN Sustainable Development Goals. My research creates **hardware-aware forecasting models** for **clean energy** (wind/load) and **urban climate resilience**, evidenced by **10+ papers (6 first-author)** and repeated SOTA gains on edge devices.

Publications & Research

Authored **10 journal articles (6 as first-author)** and **3 conference papers**, with **3 additional manuscripts under review**.

- **Physics-Informed Ensemble Learning for City-Center Grid Cell Temperature Prediction During Thermal Extremes (Urban Climate, 2025 (IF: 6.4, JCR: Q1)).**
 - Developed PIEL-NET, a hierarchical ensemble model that integrates physics-based modeling with a spatiotemporal ConvLSTM and a novel Regime Adaptive Focal Loss to predict city-core temperature by processing nine surrounding grid cells.
 - Achieved state-of-the-art performance, reducing extreme cold prediction error by 69.2% and attaining a 0.89 Dice score for early warnings, significantly outperforming benchmarks including TCN, LSTM, and iTransformer across four global cities.
- **Physics-Informed Spatio-Temporal Network with Trainable Adaptive Feature Selection for Short-Term Wind Speed Prediction. (Computers & Electrical Engineering, 2025 (IF: 4.9, JCR: Q1)).**
 - Developed PISTNet, a physics-informed spatio-temporal network featuring a Dynamic Feature Adapter for automatic feature selection, a Context-Aware Attention Mechanism, and an Adaptive Physics Penalty Loss (APPL) that selectively enforces physical constraints during training.
 - Achieved state-of-the-art performance, outperforming eight recent models (including CoReSTL, Mixformer, and iTransformer) across four global datasets, with improvements of up to 7.1% in RMSE, 8.0% in MAE, and 9.5% in SMAPE for 6-hour wind speed prediction.
- **A Shallow Hybrid Model with Dynamic Bayesian Optimisation for Wind Speed Prediction on Memory-Constrained Devices. (Computers & Electrical Engineering (IF: 4.9, JCR: Q1)).**
 - Developed D-TPE-SHM, an edge-deployable shallow hybrid that combines Independent Component Analysis with adaptive Least Absolute Shrinkage and Selection Operator for location-aware feature selection, a concurrent Long Short-Term Memory and Temporal Convolutional Network temporal module, and a Randomized Vector Functional Link attention head, while a dynamic Tree-structured Parzen Estimator tunes a bounded search space for 1 to 6 hour forecasts on Jetson Nano.
 - Achieved a state-of-the-art efficiency-accuracy trade-off on the Mongolia dataset, reducing mean squared error by 5.9% and mean absolute error by 5.0% relative to the Ensemble Deep Randomized Vector Functional Link model, and outperforming CoReSTL and iTransformer, while shrinking parameters by 38.3% with sustained on-device latency.
- **Dynamic Optimization of Recurrent Networks for Wind Prediction on Edge Devices. (IEEE Access, 2025 (IF: 3.6, JCR: Q2)).**
 - Developed an adaptive simulated annealing with memory-based rejection framework to co-optimize LSTM, GRU, and TCN architectures under memory constraints, with integrated feature selection for edge-deployment on Jetson Nano.
 - Achieved state-of-the-art edge accuracy and efficiency on the Mongolia dataset, where the optimized Gated Recurrent Unit delivered up to 22.2% lower mean squared error and 6.3% lower mean absolute percentage error than the closest baseline Flowformer, while reducing model size by 95.7% and increasing the coefficient of determination by about 2 points.

- **Under Review**

- *A Distribution Matching Framework for Zero-Shot Wind Speed Prediction on Edge Devices in Unseen Locations. (Under Review, Engineering Applications of Artificial Intelligence).*
- *PiDR-Net: A Perception-Informed Deep Recurrent Network for Short-Term Wind Speed Predictions (Submitted, Applied Soft Computing).*
- *Physics-Informed Extreme Wind Speed Prediction with a Multi-Gated Convolutional Recurrent Network VORTEX-Net (submitted to Applied Energy).*

Research Experience

My evolution into a **Full-Stack Researcher** began with six years of teaching embedded systems and signal processing courses at **iUSE**, establishing my theoretical foundations. This expertise enabled my transition to industry, where I designed and developed computer vision systems at **DLISION** and **Bond & Built**. Now at **Central South University**, my PhD research unifies these domains to propose physics-informed forecasting prediction models for sustainable systems and to validate models on edge hardware.

- **PhD Researcher – Machine Learning & AI for Sustainable Systems** | Sept 2020 – Present

- ***Conceived and executed an independent research program** from problem formulation to publication, resulting in 6 first-author papers that pioneer the integration of physical constraints with deep learning for climate and energy forecasting.*
- ***Led the full research lifecycle** for projects, identifying gaps in spatio-temporal forecasting, designing novel neural architectures, developing custom training paradigms (e.g., adaptive loss functions), and validating against state-of-the-art benchmarks across global datasets.*
- *Contributed to the successful acquisition of the **Key R&D Program of Hunan Province (Project 2020WK2007)**, aligning doctoral research with provincial strategic goals for wind energy storage and grid optimization.*
- *Established a rigorous experimental framework for **edge-AI model evaluation**, co-optimizing model architecture and hyperparameters for deployment on resource-constrained devices (NVIDIA Jetson Nano), achieving new Pareto frontiers in accuracy-efficiency trade-offs.*
- ***Mentored and collaborated** with junior PhD and master's students on model development and paper writing, fostering a productive research environment.*

- **AI/ML Consultant - Edge Computing & Computer Vision** | Bond and Built Pvt Ltd, Pakistan | July 2024 – March 2025

- *Led the **applied research and development** for a real-time footwear analytics system, tackling challenges in large-scale edge deployment and multi-stream data processing.*
- *Scaled a computer vision pipeline to process **50,000+ daily inferences across 60+ concurrent streams**, delivering actionable market intelligence and demonstrating robust system architecture.*

- **Machine Learning Engineer – Computer Vision & AI Deployment** | DLISION, Pakistan | May 2021 – Sept 2022

- *Led **computer vision projects** focused on **semantic segmentation and real-time object detection**.*
- *Optimized **UNet models** with convolutional and attention backbones, improving **segmentation accuracy by 3% against their baseline architectures** using **Focal Loss** parameters adjustment.*

- **Higher Education Research Assistant – AI for Healthcare** | International Islamic University, Pakistan | March 2019 – Feb 2020

- *Applied **Generative Adversarial Networks (GANs)** for **breast cancer detection**, improving classification accuracy.*
- *Conducted data augmentation techniques that **increased dataset diversity** and **enhanced model robustness**.*

- **Lecturer & Lab Engineer – Embedded Systems & AI** |iUSE School of Engineering and Management Sciences, Pakistan| Sept 2013 – Feb 2019
 - *Taking undergraduate courses related to **Embedded Systems, Programming, and Discrete Signal Processing**.*
 - *Designed and developed an **Instrumentation & Measurement Lab** for sensor-based data acquisition in **MATLAB**.*
 - *Led student research projects on energy management, accident alerts, and secure electronic voting.*
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Education

- PhD in Control Science & Machine Learning | Central South University, China | Sept 2020 – Present.
 - MS in Electronic Engineering| International Islamic University, Pakistan | Sept 2015 – Aug 2017.
 - BS in Electronic Engineering| International Islamic University, Pakistan | Sept 2006 – Aug 2010.
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Technical Skills

- **Machine Learning & AI:** Deep Learning | Computer Vision | Time-Series Analysis.
 - **Programming & Tools:** Python | MATLAB | C++ | TensorFlow | PyTorch | OpenCV | Edge Impulse.
 - **DevOps & Deployment:** Docker | AWS (EC2)| Git | Triton Inference Server.
 - **Hardware & IoT:** Raspberry Pi | NVIDIA Jetson | Arduino Nano BLE Sense.
 - **Data Analysis & Optimization:** Pandas | NumPy | Matplotlib | Seaborn | Feature Engineering
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Awards & Honors

- Gold Medalist – MS in Electronic Engineering | International Islamic University.
 - Hunan Provincial Government Funding for Research & Development (**Project 2020WK2007**).
 - Chinese Government CSC Scholarship for PhD Studies.
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Professional Memberships & Certifications

- Registered Engineer, Pakistan Engineering Council (ELECTRO/22837)
 - Member, Institute of Electrical and Electronics Engineers (IEEE)
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References

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